Replication and consistency

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Sample chapter for the proposed course plan.

1 Learning outcome

Explain how replication lag can produce a stale read.

Prerequisites: Lectures 03 and 05.

2 Core idea

Replication stores related state on multiple nodes. With asynchronous replication, an acknowledged update may not yet be visible on every replica. The promised read behaviour must therefore be stated explicitly.

3 Worked scenario

A registration is written to a leader. An immediate read from a lagging replica still shows the old course list.

4 Discussion

Which routing or synchronization choice could provide read-your-writes behaviour?

5 Recap

More copies do not by themselves define what a read is allowed to return.